

Day 10 – File Permissions & File Operations Challenge

Task 1: Create Files

Create empty file

```
touch devops.txt
```

Create file with content

```
echo "These are my DevOps notes" > notes.txt
```

Create script using vim

```
vim script.sh
```

Inside vim, add:

```
echo "Hello DevOps"
```

Save & exit: ESC :wq

Verify files & permissions

```
ls -l devops.txt notes.txt script.sh
```

📷 Screenshot: file list with permissions

Task 2: Read Files

```
cat notes.txt
```

```
vim -R script.sh
```

```
head -n 5 /etc/passwd
```

```
tail -n 5 /etc/passwd
```

📷 Screenshot: outputs

Task 3: Understand Permissions

Check permissions

```
ls -l devops.txt notes.txt script.sh
```

Example explanation

- -rw-r--r--
 - Owner: read, write
 - Group: read
 - Others: read
- -rw-rw-r--
 - Owner & group can write
 - Others read-only

🔑 Key idea:

Permissions are always **Owner | Group | Others**

Task 4: Modify Permissions

Make script executable

```
chmod +x script.sh
./script.sh
```

Expected output:

Hello DevOps

Make devops.txt read-only

```
chmod a-w devops.txt
ls -l devops.txt
```

Set notes.txt to 640

```
chmod 640 notes.txt
ls -l notes.txt
```

Create directory with 755

```
mkdir project
chmod 755 project
ls -ld project
```

📷 Screenshot: permission changes

Task 5: Test Permissions

Try writing to read-only file

```
echo "test" >> devops.txt
```

Expected error:

Permission denied

Try executing without execute permission

```
chmod -x script.sh  
./script.sh
```

Expected error:

Permission denied

 Screenshot: error messages

Documentation File

Create **day-10-file-permissions.md**

Day 10 Challenge – File Permissions & Operations

Files Created

- devops.txt
- notes.txt
- script.sh
- project/

Permission Changes

- script.sh → made executable with `chmod +x`
- devops.txt → read-only for all users
- notes.txt → set to 640
- project/ → directory permission 755

Commands Used

- touch, echo, vim
- cat, head, tail
- chmod, ls -l, ls -ld

What I Learned

- How Linux permissions control file access
- Numeric vs symbolic chmod
- Why execute permission is required to run scripts